

Math 104C Project 2: Numerical Methods for Partial Differential Equations

Jiajia Liu
Perm ID: A0J2E85

Code link: <https://github.com/samanthaliujiajia/math104c-project2>

1 Introduction

This project investigates finite-difference methods for elliptic and parabolic partial differential equations. The experiments compare accuracy, stability, mesh refinement, and computational behavior for the five-point finite-difference method, Forward Difference, Backward Difference, and Crank–Nicolson.

The elliptic experiments use square grids with $h = 1/4$, $h = 1/8$, and $h = 1/16$. The heat equation experiments use multiple choices of h and k , including both stable and unstable Forward-Difference cases. Whenever an exact solution is available, the absolute error is computed at mesh points.

2 Numerical Methods

For elliptic equations on the unit square, the unknowns are the interior grid values w_{ij} . The boundary values are known from the problem statement and are moved to the right-hand side of the linear system. At each interior point, the five-point stencil approximates

$$u_{xx}(x_i, y_j) + u_{yy}(x_i, y_j) \approx \frac{w_{i+1,j} + w_{i-1,j} + w_{i,j+1} + w_{i,j-1} - 4w_{i,j}}{h^2}.$$

For the Laplace problem this gives

$$w_{i+1,j} + w_{i-1,j} + w_{i,j+1} + w_{i,j-1} - 4w_{ij} = 0,$$

while for the Poisson experiment the right-hand side is $h^2 f(x_i, y_j)$. The coefficient matrix is sparse because each interior point only interacts with its four nearest neighbors.

I solve the elliptic systems in two ways. The sparse direct solver is used as a baseline because it solves the matrix equation $Aw = b$ in one algebraic step. I also use successive over-relaxation (SOR), which updates the approximate solution point by point:

$$w_{ij}^{(m+1)} = (1 - \omega)w_{ij}^{(m)} + \frac{\omega}{4} (w_{i+1,j} + w_{i-1,j} + w_{i,j+1} + w_{i,j-1} - h^2 f_{ij}).$$

In the experiments I use $\omega = 1.6$ and stop when the residual is below the chosen tolerance. Reporting the iteration count helps compare accuracy with computational effort.

For the heat equation $u_t = u_{xx}$, the Forward-Difference method is

$$w_{i,j+1} = \lambda w_{i-1,j} + (1 - 2\lambda)w_{i,j} + \lambda w_{i+1,j}, \quad \lambda = \frac{k}{h^2}.$$

This method is explicit, so each new time level is computed directly from the previous one. Its simplicity is useful, but it is only conditionally stable for $\lambda \leq 1/2$.

The Backward-Difference method uses the spatial second difference at the new time level:

$$-\lambda w_{i-1,j+1} + (1 + 2\lambda)w_{i,j+1} - \lambda w_{i+1,j+1} = w_{i,j}.$$

This produces a tridiagonal linear system at each time step. It is first-order accurate in time, but its implicit form is unconditionally stable for the heat equation.

Crank–Nicolson averages the spatial second difference between the old and new time levels. In matrix form it can be written as

$$\left(I - \frac{\lambda}{2}A\right)w^{j+1} = \left(I + \frac{\lambda}{2}A\right)w^j,$$

where A is the centered second-difference matrix. This method is also implicit, but it has second-order accuracy in time. Comparing these three heat methods makes the tradeoff between explicit simplicity, implicit stability, and temporal accuracy visible in the numerical results.

3 Problem A: Elliptic PDE

Problem A solves

$$u_{xx} + u_{yy} = 0, \quad 0 < x < 1, \quad 0 < y < 1,$$

with boundary conditions

$$u(0, y) = 0, \quad u(x, 0) = 0, \quad u(x, 1) = 100x, \quad u(1, y) = 100y.$$

The exact solution is $u(x, y) = 100xy$, which is bilinear. Since the five-point stencil is exact for this function up to roundoff, this problem is a useful check that boundary values and matrix assembly are correct.

Table 1: Elliptic PDE error summary and SOR iteration counts.

problem	method	h	max error	iterations	residual
A	Direct	0.250000	7.105e-15	0	2.842e-14
A	SOR omega=1.6	0.250000	5.321e-11	56	1.873e-10
A	Direct	0.125000	2.132e-14	0	3.553e-14
A	SOR omega=1.6	0.125000	6.425e-11	58	1.914e-10
A	Direct	0.062500	2.132e-14	0	8.527e-14
A	SOR omega=1.6	0.062500	8.285e-10	137	9.442e-11
C1	Direct	0.250000	0.053029	0	8.882e-16
C1	SOR omega=1.6	0.250000	0.053029	47	1.591e-10
C1	Direct	0.125000	0.012951	0	6.384e-16
C1	SOR omega=1.6	0.125000	0.012951	48	8.876e-11
C1	Direct	0.062500	0.003219	0	1.527e-15
C1	SOR omega=1.6	0.062500	0.003219	114	8.457e-11

Table 2: Observed orders for elliptic experiments.

problem	method	observed order
A	Direct	-0.792481
A	SOR omega=1.6	-1.980335
C1	Direct	2.021060
C1	SOR omega=1.6	2.021061

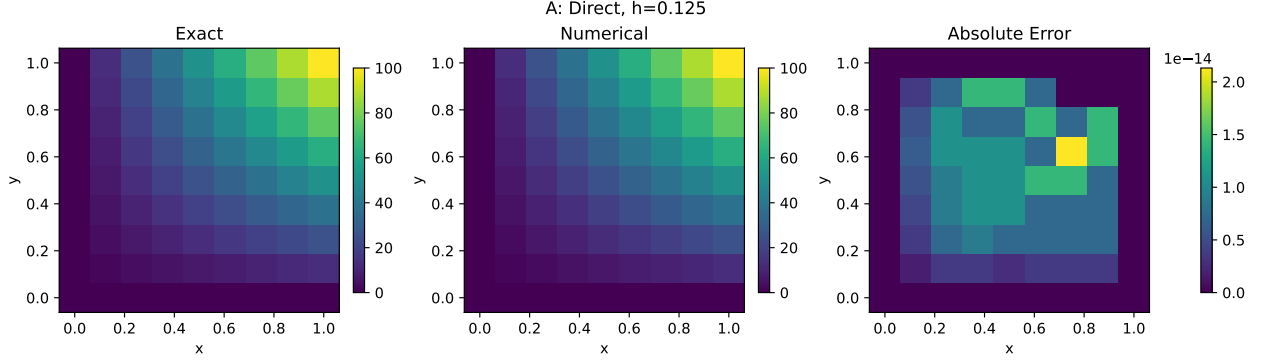


Figure 1: Problem A exact solution, numerical solution, and absolute error for $h = 1/8$.

4 Problem B: Heat Equation

Problem B solves

$$u_t = u_{xx}, \quad 0 < x < 1, \quad t > 0,$$

with $u(0, t) = u(1, t) = 0$ and $u(x, 0) = \sin(\pi x)$. The exact solution is

$$u(x, t) = e^{-\pi^2 t} \sin(\pi x).$$

The Forward-Difference experiments include $\lambda = 0.4$, which satisfies the stability condition, and $\lambda = 0.6$, which violates it.

Table 3: Heat equation error summary.

problem	case	method	h	k	λ	final error	max error
B	B_stable_h0.1	Forward Difference	0.100000	0.004000	0.400000	0.004294	0.004294
B	B_stable_h0.1	Backward Difference	0.100000	0.004000	0.400000	0.010112	0.010112
B	B_stable_h0.1	Crank-Nicolson	0.100000	0.004000	0.400000	0.002981	0.002981
B	B_refined_h0.05	Forward Difference	0.050000	0.001000	0.400000	0.001063	0.001063
B	B_refined_h0.05	Backward Difference	0.050000	0.001000	0.400000	0.002561	0.002561
B	B_refined_h0.05	Crank-Nicolson	0.050000	0.001000	0.400000	7.535e-04	7.535e-04
B	B_unstable_h0.1	Forward Difference	0.100000	0.010000	1.000000	0.015756	0.015756
B	B_unstable_h0.1	Backward Difference	0.100000	0.010000	1.000000	0.020320	0.020320
B	B_unstable_h0.1	Crank-Nicolson	0.100000	0.010000	1.000000	0.002734	0.002734
C2	C2_stable_h0.1	Forward Difference	0.100000	0.004000	0.400000	0.003285	0.017047
C2	C2_stable_h0.1	Backward Difference	0.100000	0.004000	0.400000	0.008843	0.036656
C2	C2_stable_h0.1	Crank-Nicolson	0.100000	0.004000	0.400000	0.002355	0.010840
C2	C2_refined_h0.05	Forward Difference	0.050000	0.001000	0.400000	8.740e-04	0.004294
C2	C2_refined_h0.05	Backward Difference	0.050000	0.001000	0.400000	0.002181	0.010114
C2	C2_refined_h0.05	Crank-Nicolson	0.050000	0.001000	0.400000	6.247e-04	0.002981

Table 4: Observed orders for stable heat equation refinement cases.

problem	method	observed order
B	Backward Difference	1.981501
B	Crank-Nicolson	1.983931
B	Forward Difference	2.014890
C2	Backward Difference	1.857639
C2	Crank-Nicolson	1.862650
C2	Forward Difference	1.989060

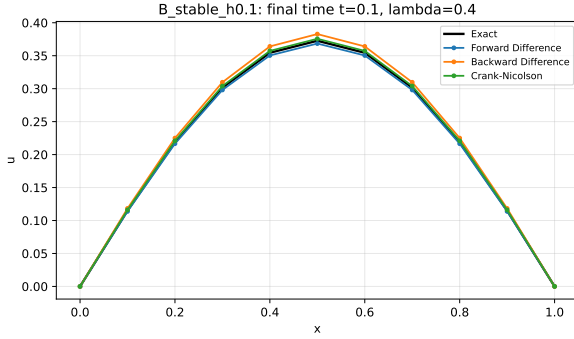
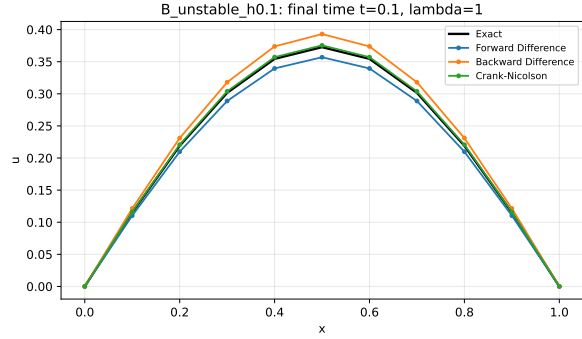
Stable Forward case, $\lambda = 0.4$.Unstable Forward case, $\lambda = 0.6$.

Figure 2: Problem B final-time solution profiles.

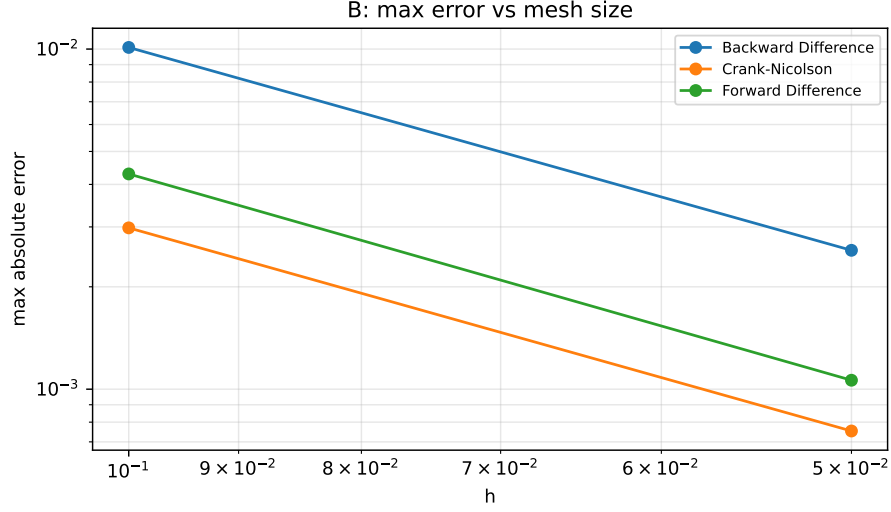


Figure 3: Problem B max error under stable mesh refinement.

5 Problem C: Additional Experiments

The first additional experiment solves a Poisson equation with a manufactured solution:

$$u_{xx} + u_{yy} = -2\pi^2 \sin(\pi x) \sin(\pi y), \quad u = 0 \text{ on the boundary.}$$

The exact solution is $u(x, y) = \sin(\pi x) \sin(\pi y)$. Unlike Problem A, this solution is not bilinear, so the second-order spatial truncation error is visible in the grid refinement results.

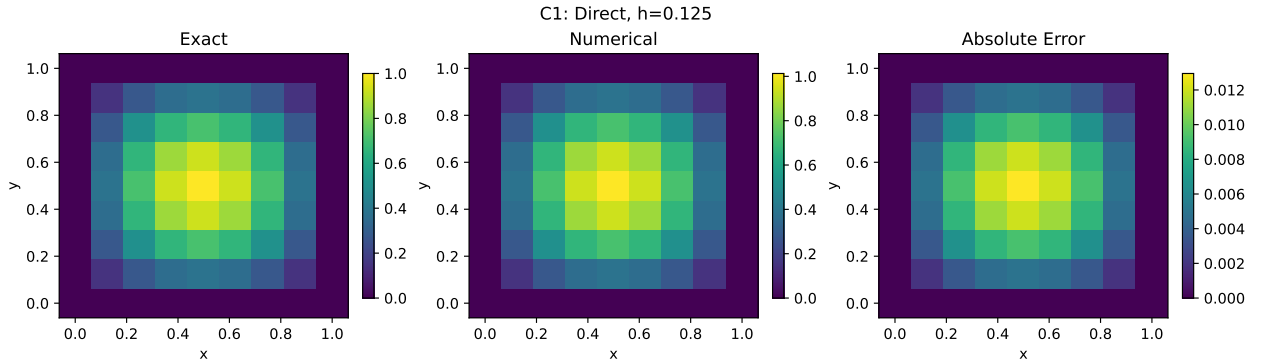


Figure 4: Poisson manufactured-solution experiment for $h = 1/8$.

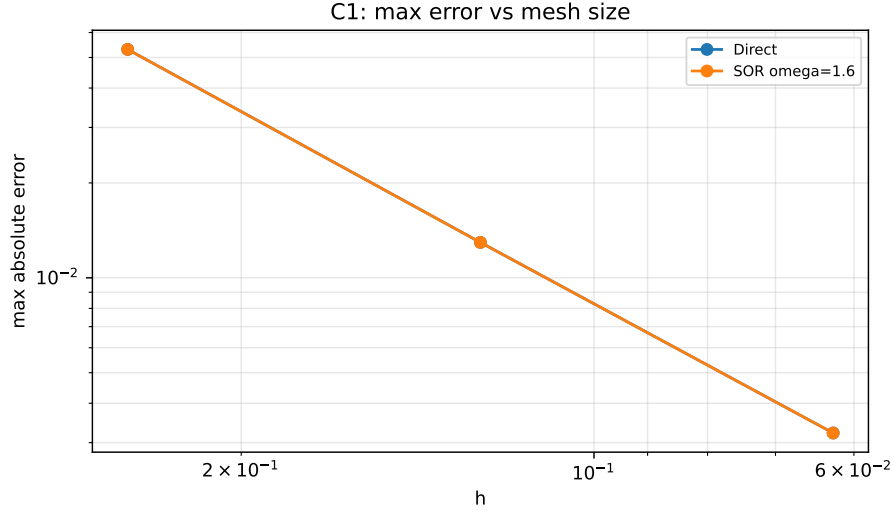


Figure 5: Problem C1 max error under mesh refinement.

The second additional experiment changes the heat equation initial condition to

$$u(x, 0) = \sin(2\pi x),$$

whose exact solution is

$$u(x, t) = e^{-4\pi^2 t} \sin(2\pi x).$$

This higher-frequency mode decays faster and is more demanding for a coarse mesh.

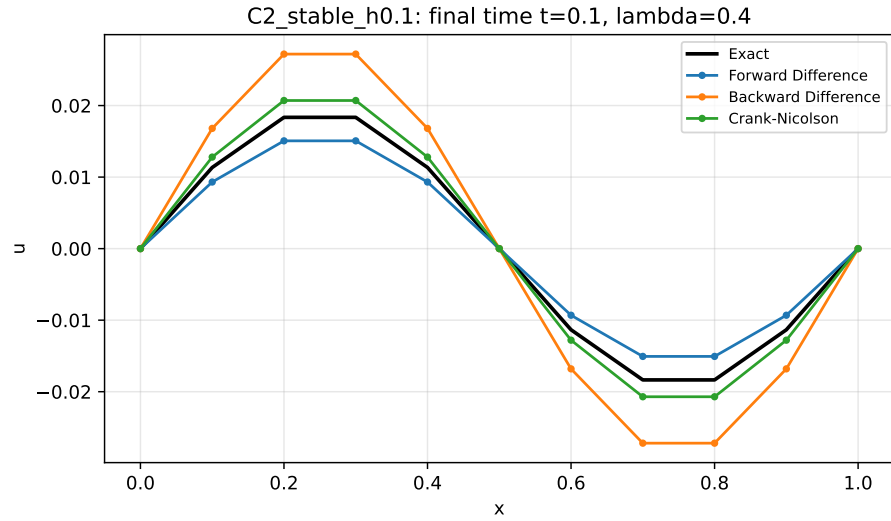


Figure 6: Problem C2 final-time profiles for the different initial condition.

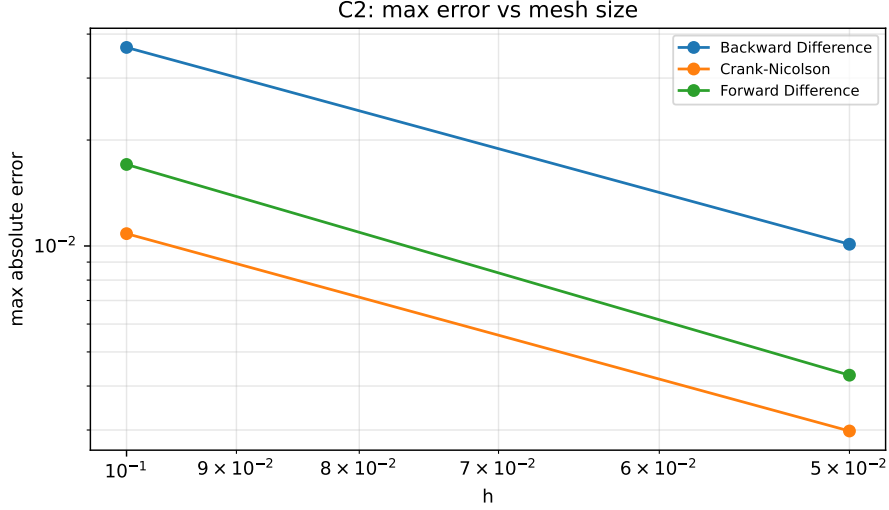


Figure 7: Problem C2 max error under stable mesh refinement.

6 Discussion and Analysis

Problem A confirms both the finite-difference stencil and the treatment of boundary data. Since the exact solution $u(x, y) = 100xy$ is bilinear, the second derivatives are represented exactly by the centered difference formula up to roundoff error. Therefore, the small errors in the table are not surprising: they mainly measure floating-point error and the accuracy of the linear solve, not truncation error. This is still an important test because an incorrect sign in the matrix or a missing boundary contribution would immediately destroy the linear solution.

The direct solver and SOR solver give essentially the same numerical solution, but their computational behavior is different. The sparse direct solver treats the linear system as one algebraic problem. SOR updates the grid values repeatedly, so the number of iterations grows as the mesh is refined. This is a typical feature of stationary iterative methods: finer grids have more unknowns, and information moves across the domain gradually through local updates. The residuals show that the iteration was stopped only after the algebraic system was solved to a tight tolerance.

Problem B illustrates the central stability difference between explicit and implicit time stepping. For the heat equation, the Forward-Difference method is consistent with local truncation error $O(k + h^2)$, but consistency alone does not guarantee useful numerical results. The method also needs $\lambda = k/h^2 \leq 1/2$. In the stable case, the final-time profiles remain smooth and the error decreases under refinement. In the unstable case, the method amplifies high-frequency error modes instead of damping them, so the computed solution can develop nonphysical oscillations even though the exact heat equation smooths and decays the solution.

Backward Difference and Crank–Nicolson behave differently because the new time level is included implicitly. Backward Difference is more diffusive, which can make it slightly less accurate for the same mesh, but that diffusion is also why it strongly damps unstable modes. Crank–Nicolson is less artificially dissipative and has second-order temporal accuracy, so it usually gives a better balance between accuracy and stability. The experiments show this tradeoff: implicit methods require

solving a linear system at every time step, but they allow larger k without the Forward-Difference stability failure.

The Poisson manufactured-solution experiment is a better test of spatial convergence than Problem A because $u = \sin(\pi x) \sin(\pi y)$ is not exactly represented by the five-point stencil. The observed errors decrease as the mesh is refined, which matches the theoretical second-order behavior of the centered spatial approximation. This experiment also shows why manufactured solutions are useful: the source term is chosen so the exact solution is known, making it possible to measure the numerical error directly.

The second additional experiment changes the heat equation initial condition to $\sin(2\pi x)$. This mode has a higher spatial frequency and decays like $e^{-4\pi^2 t}$, much faster than the $\sin(\pi x)$ mode. Numerically, this means a coarse mesh has fewer points per wavelength, so spatial resolution matters more. The experiment connects error propagation with the structure of the PDE: the heat equation damps high-frequency physical modes, but a numerical method can still introduce additional error if the grid or time step is too coarse.

Overall, the experiments show the relationship between consistency, stability, and convergence. The finite-difference formulas are consistent approximations, but stability determines whether the accumulated error remains controlled. For elliptic problems, the main computational issue is solving the sparse linear system accurately and efficiently. For parabolic problems, the choice of time step is just as important as the spatial mesh, especially for explicit methods. Thus the best method depends not only on formal order, but also on stability requirements and computational cost.

7 Conclusion

The experiments support the theoretical results for finite-difference PDE methods. The five-point elliptic method is accurate and has a sparse matrix structure that can be solved either directly or iteratively. For parabolic problems, consistency alone is not enough: the Forward-Difference method also requires the stability condition $\lambda \leq 1/2$. Backward Difference and Crank–Nicolson are more stable for large time steps, with Crank–Nicolson offering higher temporal accuracy.